

-
8. Which two statements prove that "Senait is a child"? A. Senait lives with her parents. B. Senait is the little sister of Bekele C. Bekele has just turned 13. D. Bekele is still in school. (1)
- A. C and D
B. A and C
C. B and D
D. B and C
-
9. Which two statements prove that Eleni is a doctor? A. Eleni loves people. B. Eleni's father is teacher. C. Eleni's brother is a doctor. D. Eleni has the same job as her brother. (1)
- A. C and D
B. A and D
C. B and C
D. A and C
-
10. *Regular physical activity provided numerous health benefits. : from leaner bodies and ,lower blood pressure to improved mental health and cognitive functioning. As the school physical education program promotes physical activity and can teach skills as well as form or change behavior, it holds an important key to influencing health and well being across life span. To improve the fitness of students, we need to rethink the design and delivery of school- based physical education program. Adults in some countries think that information about health was more important for students to learn than content in language arts, mathematics, science, history or any other subject. Despite this high ranking, most schools devote minimum curriculum time to teaching students how to lead healthy lives. Our first step might be to consider ways to increase curriculum time devoted to physical education. In addition, schools need to thoughtfully analyze the design and delivery of school physical education programs to ensure that they are engaging, developmentally appropriate, inclusive and instructionally powerful. Adapted from Test Prep Review: A resource from Mometrix* (1)
- According to this passage, regular physical activity is needed to
- A. lose weight.
B. control one's blood pressure.
C. improve ones' physical as well as mental health.
D. improve ones' cognitive skills.
-
11. *Regular physical activity provided numerous health benefits. : from leaner bodies and ,lower blood pressure to improved mental health and cognitive functioning. As the school physical education program promotes physical activity and can teach skills as well as form or change behavior, it holds an important key to influencing health and well being across life span. To improve the fitness of students, we need to rethink the design and delivery of school- based physical education program. Adults in some countries think that information about health was more important for students to learn than content in language arts, mathematics, science, history or any other subject. Despite this high ranking, most schools devote minimum curriculum time to teaching students how to lead healthy lives. Our first step might be to consider ways to increase curriculum time devoted to physical education. In addition, schools need to thoughtfully analyze the design and delivery of school physical education programs to ensure that they are engaging, developmentally appropriate, inclusive and instructionally powerful. Adapted from Test Prep Review: A resource from Mometrix* (1)
- In order to tone up the physical education program,
- A. an assessment of the existing program should be made.
B. it should be compulsory at school.
C. the program should be reoriented and implemented.
D. a committee should be set up in every school.

12. Regular physical activity provided numerous health benefits : from leaner bodies and ,lower blood pressure to improved mental health and cognitive functioning. As the school physical education program promotes physical activity and can teach skills as well as form or change behavior, it holds an important key to influencing health and well being across life span. To improve the fitness of students, we need to rethink the design and delivery of school- based physical education program. Adults in some countries think that information about health was more important for students to learn than content in language arts, mathematics, science, history or any other subject. Despite this high ranking, most skills devote minimum curriculum time to teaching students how to lead healthy lives. Our first step might be to consider ways to increase curriculum time devoted to physical education. In addition, schools need to thoughtfully analyze the design and delivery of school physical education programs to ensure that they are engaging, developmentally appropriate, inclusive and instructionally powerful. Adapted from Test Prep Review: A resource from Mometrix
- According to some countries, health education is more important than teaching
- (1)

- A.** natural sciences. **B.** social sciences.
C. liberal arts. **D.** any subject.

13. *Regular physical activity provided numerous health benefits. : from leaner bodies and ,lower blood pressure to improved mental health and cognitive functioning. As the school physical education program promotes physical activity and can teach skills as well as form or change behavior, it holds an important key to influencing health and well being across life span. To improve the fitness of students, we need to rethink the design and delivery of school- based physical education program. Adults in some countries think that information about health was more important for students to learn than content in language arts, mathematics, science, history or any other subject. Despite this high ranking, most schools devote minimum curriculum time to teaching students how to lead healthy lives. Our first step might be to consider ways to increase curriculum time devoted to physical education. In addition, schools need to thoughtfully analyze the design and delivery of school physical education programs to ensure that they are engaging, developmentally appropriate, inclusive and instructionally powerful. Adapted from Test Prep Review: A resource from Mometrix*
- The author wants the reoriented physical education program to be

- A. comprehensive.
- B. short.
- C. thoughtful.
- D. relevant.

14. *Marie Curie was one of the most accomplished scientists in history. Together with her husband, Pierre, she discovered radium, an element widely used for treating cancer, and studied uranium and other radioactive substances. Pierre and Marie's amicable collaboration later helped to unlock the secrets of the atom. Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics. Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists of her day, one of whom was Pierre Curie. Marie and Pierre were married in 1895 and spent many productive years working together in the physics laboratory. A short time after they discovered radium, Pierre was killed by a horse-drawn wagon in 1906. Marie was stunned by this horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the joy that they had shared in scientific research. The fact that she had two young daughters to raise by herself greatly increased her distress. Curie's feeling of desolation finally began to fade when she was asked to succeed her husband as a physics professor at the Sorbonne. She was the first woman to be given a professorship at the world-famous university. In 1911 she received the Nobel Prize in chemistry for isolating radium. Although Marie Curie eventually suffered a fatal illness from her long exposure to radium, she never became disillusioned about her work. Regardless of the consequences, she had dedicated herself to science and to revealing the mysteries of the physical world. Adapted from Test Prep Review: A resource from Mometrix* (1)
- When she learned that she could not attend the university in Warsaw, she felt _____.
- A. annoyed
B. hopeless
C. worried
D. delighted

15. *Marie Curie was one of the most accomplished scientists in history. Together with her husband, Pierre, she discovered radium, an element widely used for treating cancer, and studied uranium and other radioactive substances. Pierre and Marie's amicable collaboration later helped to unlock the secrets of the atom. Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics. Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists of her day, one of whom was Pierre Curie. Marie and Pierre were married in 1895 and spent many productive years working together in the physics laboratory. A short time after they discovered radium, Pierre was killed by a horse-drawn wagon in 1906. Marie was stunned by this horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the joy that they had shared in scientific research. The fact that she had two young daughters to raise by herself greatly increased her distress. Curie's feeling of desolation finally began to fade when she was asked to succeed her husband as a physics professor at the Sorbonne. She was the first woman to be given a professorship at the world-famous university. In 1911 she received the Nobel Prize in chemistry for isolating radium. Although Marie Curie eventually suffered a fatal illness from her long exposure to radium, she never became disillusioned about her work. Regardless of the consequences, she had dedicated herself to science and to revealing the mysteries of the physical world. Adapted from Test Prep Review: A resource from Mometrix* (1)
- Marie _____ by leaving Poland and traveling to France to enter the Sorbonne.
- A. showed intelligence
B. challenged authority
C. was distressed
D. behaved

16. Marie Curie was one of the most accomplished scientists in history. Together with her husband, Pierre, she discovered radium, an element widely used for treating cancer, and studied uranium and other radioactive substances. Pierre and Marie's amicable collaboration later helped to unlock the secrets of the atom. Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics. Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists of her day, one of whom was Pierre Curie. Marie and Pierre were married in 1895 and spent many productive years working together in the physics laboratory. A short time after they discovered radium, Pierre was killed by a horse-drawn wagon in 1906. Marie was stunned by this horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the joy that they had shared in scientific research. The fact that she had two young daughters to raise by herself greatly increased her distress. Curie's feeling of desolation finally began to fade when she was asked to succeed her husband as a physics professor at the Sorbonne. She was the first woman to be given a professorship at the world-famous university. In 1911 she received the Nobel Prize in chemistry for isolating radium. Although Marie Curie eventually suffered a fatal illness from her long exposure to radium, she never became disillusioned about her work. Regardless of the consequences, she had dedicated herself to science and to revealing the mysteries of the physical world. Adapted from Test Prep Review: A resource from Mometrix
- _____ she remembered their joy together.
- A. Worriedly
B. Dejectedly
C. Happily
D. Tearfully
-
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- Her _____ began to fade when she returned to the Sorbonne to succeed her husband.
- A. anger
B. misfortune
C. disappointment
D. wretchedness
-
18. Which continuation is right? I don't trust social medias. _____.
- A. I have never and I will never
B. I never have and I never will
C. Never can I and never I will
D. Never I have and never I will

19. If I were in your position, I _____ (1)
- A. Will never have done such silly things. B. Would never have done such silly things.
C. Would have done such silly things. D. would never do such silly things.
20. She contracted COVID 19 easily because she was obese and had other preexisting diseases. The doctor said (1)
that if she _____ obese, her chances of getting the disease would have been slim.
- A. Had been B. had not been
C. were not D. were
21. I: Haven't you seen this film? _____ (1)
- A. No, I have B. Yes, I have
C. Yes, have I D. Yes, I haven't
22. The man vehemently denied all the charges of killing that were brought against him. (1)
- A. hysterically B. forcefully
C. devoutly D. carelessly
23. Which word means the same as "counterfeit?" (1)
- A. robbery B. fake
C. copy D. match
24. He inherited a large amount of fortune from his parents. So, he lives a very lavish lifestyle. (1)
- A. stingy B. extravagant
C. spartan D. humble
25. He has reached across the world, fomenting violence against the west. (1)
- A. instigating B. planning
C. subduing D. negating
26. Which of the following words cannot go together with the others? (1)
- A. Wind B. Sky
C. Rain D. Snow
27. EAGLE is to BIRD as ROSE is to _____ (1)
- A. BEAUTIFUL B. GENTLEMAN
C. FLOWER D. ANIMAL
28. CAR is to TRANSPORTATION as GAME is to _____ (1)
- A. INFORMATION B. ENTERTAINMENT
C. FOOTBALL D. MANIPULATION
29. SPEED is to FAST as TEMPERATURE is to _____ (1)
- A. ICE B. THERMOMETER
C. CLIMATE D. WARM
30. FAST is to RUNNING as SLOW is to _____ (1)
- A. LAZY B. PACE
C. WALKING D. SPEED

-
- 31. CANDLE: WAX** (1)
- | | |
|-----------------|----------------------|
| A. TREE: LEAVES | B. BRANCH: WOOD |
| C. PAPER: PULP | D. BAMBOO: FURNITURE |
-
- 32. CHEF: FOOD** (1)
- | | |
|---------------------|--------------------|
| A. BRUSH: WALL | B. PAINT: PAINTER |
| C. BRICK: CARPENTER | D. SCULPTOR: STONE |
-
- 33. BOARD: TRAIN** (1)
- | | |
|-----------------|-------------------|
| A. WALK: SHOE | B. CHAIR: SIT |
| C. MOUNT: HORSE | D. BICYCLE: DRIVE |
-
- 34. The Triangular Number Sequence T_n can be defined recursively as $T_1 = 1, T_n = T_{n-1} + n$ for $n > 1$. What is the eleventh term of the sequence?** (1)
- | | |
|-------|-------|
| A. 66 | B. 45 |
| C. 78 | D. 55 |
-
- 35. The next term of the sequence 162, 54, 18, 6, . . . equal to** (1)
- | | |
|--------|------|
| A. 2 | B. 1 |
| C. 1/3 | D. 3 |
-
- 36. The first four terms of an arithmetic sequence are 39, 33, 27 and 21. What is the tenth term of the sequence?** (1)
- | | |
|--------|--------|
| A. -21 | B. -15 |
| C. -33 | D. -27 |
-
- 37. The sum of the first n terms of an arithmetic progression whose first term is $2n^2 + n$, then what is the tenth term of the sequence?** (1)
- | | |
|-------|-------|
| A. 39 | B. 40 |
| C. 38 | D. 41 |
-
- 38. For Questions 40- 45 Use The Following Information. The following table gives the expenditure (in millions of Birr) of a certain company per year over five years. Items of Expenditure Year | Salary | Fuel and Transport | Bonuses | Interest on Loans | Taxes 2008 | 288 | 98 | 3.1 | 23.4 | 83 2009 | 342 | 112 | 2.52 | 32.5 | 108 2010 | 324 | 101 | 3.84 | 41.2 | 74 2011 | 336 | 133 | 3.68 | 36.4 | 88 2012 | 422 | 142 | 3.96 | 49.4 | 98** (1)
- What is the average amount of interest on loans per year that the company has to pay during the five year period?
- | | |
|----------|----------|
| A. 33.72 | B. 33.43 |
| C. 36.58 | D. 34.18 |
-
- 39. For Questions 40- 45 Use The Following Information. The following table gives the expenditure (in millions of Birr) of a certain company per year over five years. Items of Expenditure Year | Salary | Fuel and Transport | Bonuses | Interest on Loans | Taxes 2008 | 288 | 98 | 3.1 | 23.4 | 83 2009 | 342 | 112 | 2.52 | 32.5 | 108 2010 | 324 | 101 | 3.84 | 41.2 | 74 2011 | 336 | 133 | 3.68 | 36.4 | 88 2012 | 422 | 142 | 3.96 | 49.4 | 98** (1)
- What percent of the total amount of salary was paid as bonuses by the company during the given period?
- | | |
|----------|---------|
| A. 0.5% | B. 0.1% |
| C. 1.25% | D. 1.0% |

40. For Questions 40- 45 Use The Following Information. The following table gives the expenditure (in millions of Birr) of a certain company per year over five years. Items of Expenditure Year | Salary | Fuel and Transport | Bonuses | Interest on Loans | Taxes 2008 | 288 | 98 | 3.1 | 23.4 | 83 2009 | 342 | 112 | 2.52 | 32.5 | 108 2010 | 324 | 101 | 3.84 | 41.2 | 74 2011 | 336 | 133 | 3.68 | 36.4 | 88 2012 | 422 | 142 | 3.96 | 49.4 | 98 (1)

In which year did the company have the highest expenditure?

- A. 2009
B. 2008
C. 2010
D. 2012

41. For Questions 40- 45 Use The Following Information. The following table gives the expenditure (in millions of Birr) of a certain company per year over five years. Items of Expenditure Year | Salary | Fuel and Transport | Bonuses | Interest on Loans | Taxes 2008 | 288 | 98 | 3.1 | 23.4 | 83 2009 | 342 | 112 | 2.52 | 32.5 | 108 2010 | 324 | 101 | 3.84 | 41.2 | 74 2011 | 336 | 133 | 3.68 | 36.4 | 88 2012 | 422 | 142 | 3.96 | 49.4 | 98 (1)

What is the total expenditure of the company in the year 2012?

- A. 715.36
B. 1710
C. 456.36
D. 586

42. For Questions 40- 45 Use The Following Information. The following table gives the expenditure (in millions of Birr) of a certain company per year over five years. Items of Expenditure Year | Salary | Fuel and Transport | Bonuses | Interest on Loans | Taxes 2008 | 288 | 98 | 3.1 | 23.4 | 83 2009 | 342 | 112 | 2.52 | 32.5 | 108 2010 | 324 | 101 | 3.84 | 41.2 | 74 2011 | 336 | 133 | 3.68 | 36.4 | 88 2012 | 422 | 142 | 3.96 | 49.4 | 98 (1)

For What item or purpose was the company's expenditure the least during the given period?

- A. Bonus
B. Salary
C. Taxes
D. Transport and Fuel

43. If $x = 15\%$ of 30 and $y = 30\%$ of 15, then which of the following is true? (1)

- A. x and y are equal
B. x is less than y
C. The relation between x and y cannot be determined
D. y is less than x

44. Student X and student Y have 2000 Birr. If $\frac{4}{15}$ of X's share is equal to $\frac{2}{5}$ of Y's share, how much amount does Y have? (1)

- A. 800 Birr
B. 1500 Birr
C. 1200 Birr
D. 1100 Birr

45. What is the simplified form of $(2x + 3)(x + 6) - (2x - 5)(x + 10)$? (1)

- A. 16
B. 32
C. 72
D. 68

46. If the square of a number increased by two times the number is equal to six times the number increased by twelve, which one of the following is the possible value of the number? (1)

- A. 4
B. 6
C. -4
D. -6

47. The monthly telephone bill of Getachew was Birr 400 and it increases by 10% after 12 months, and another 20% increase is applied six months later. What is the price of Getachew's telephone bill after 18 months? (1)

- A. 520 Birr
B. 528 Birr
C. 489 Birr
D. 430 Birr

-
48. The sum of the ages of the father and his son is equal to sixty years. Six years ago the age of the father was five times the age of the son. What will be the age of the son six years from now? (1)
- A. 19 years
B. 23 years
C. 22 years
D. 20 years
-
49. What is the sum of the following infinite series $S = 2 - \frac{1}{3} + 1 - \frac{1}{9} + \frac{1}{2} - \frac{1}{27} + \frac{1}{4} - \frac{1}{81} + \dots$? (1)
- A. $\frac{7}{2}$
B. $\frac{32}{9}$
C. $\frac{41}{9}$
D. $\frac{9}{2}$
-
50. 1.2 (II) If $(1^2 + 2^2 + 3^2 + \dots + 10^2) = a$, then what is the value of $(2^2 + 4^2 + 6^2 + \dots + 20^2)$ in terms of a ? (1)
- A. $4a$
B. $2a$
C. $a+4$
D. $a+2$
-
51. If $y > 2$, compare the following two quantities X and Y whose information are given below. Quantity X: $\frac{3y-2}{5}$ (1)
Quantity Y: y
- A. Quantity Y is greater than Quantity X.
B. Quantity X is greater than Quantity Y.
C. The relation cannot be determined with the information provided.
D. The two quantities are equal.
-
52. What is the probability of getting exactly two tails when three coins are tossed at the same time? (1)
- A. $\frac{1}{4}$
B. $\frac{1}{8}$
C. $\frac{3}{4}$
D. $\frac{3}{8}$
-
53. What is the probability that a number selected at random from the first 50 natural numbers is prime and odd number? (1)
- A. $\frac{3}{10}$
B. $\frac{7}{25}$
C. $\frac{8}{25}$
D. $\frac{3}{5}$
-
54. If $3x + 7y = 80$ and $x - y = 10$, then what is the value of $x + y$? (1)
- A. 10
B. 15
C. 25
D. 20

55. Which of the following system of linear inequalities is represented by the graph given below?

(1)

[Figure: A Cartesian plane with the horizontal x-axis (arrowhead at the right, labelled x) and the vertical y-axis (arrowhead at the top, labelled y). Three straight lines are drawn, all as plain solid lines with no shading of any region: (1) a line of positive slope rising left to right, crossing the x-axis at the marked point -1 and the y-axis at the marked point 1, so it passes through (-1, 0) and (0, 1); (2) a line of negative slope falling left to right, crossing the y-axis at the same point 1 and the x-axis at the marked point 1, so it passes through (0, 1) and (1, 0); the two lines cross each other on the y-axis at (0, 1). (3) a horizontal line running the full width of the drawing at the level of the marked point -1 on the y-axis, that is the line $y = -1$. The three lines enclose a triangle whose apex is at (0, 1) and whose base lies on $y = -1$. Axis marks shown and labelled: -1 and 1 on the x-axis, 1 and -1 on the y-axis. No region is shaded and no arrows indicate a solution side.]

A.
$$\begin{cases} y \leq x + 1 \\ y \leq x - 1 \\ y \leq -1 \end{cases}$$

B.
$$\begin{cases} y \leq x + 1 \\ y \leq -x + 1 \\ y \leq -1 \end{cases}$$

C.
$$\begin{cases} y \leq x + 1 \\ y \leq x - 1 \\ y \geq -1 \end{cases}$$

D.
$$\begin{cases} y \leq x + 1 \\ y \leq -x + 1 \\ y \geq -1 \end{cases}$$

56. The square of the 7th term of an arithmetic progression with positive common difference is equal to the product of the 3rd and 17th terms, then what is the ratio of the first term to the common difference? (1)

A. 3 : 4

B. 2 : 3

C. 4 : 3

D. 3 : 2

— END OF EXAMINATION —